Phase 5 — Monitoring, alarms, and budget alerts

CloudWatch collects operational metrics and evaluates alarms. The architecture reference shows alarms for CPU, ALB target health, and RDS CPU, with notifications published to the SNS topic CloudWatch-alarms-topic. AWS Budgets provides spending alerts.

Why this phase matters

Monitoring highlights unhealthy compute, database, and load-balancer resources. SNS distributes alert messages to the confirmed subscriber. Budget notifications provide early warning; they do not automatically stop resources or create a hard spending cap.

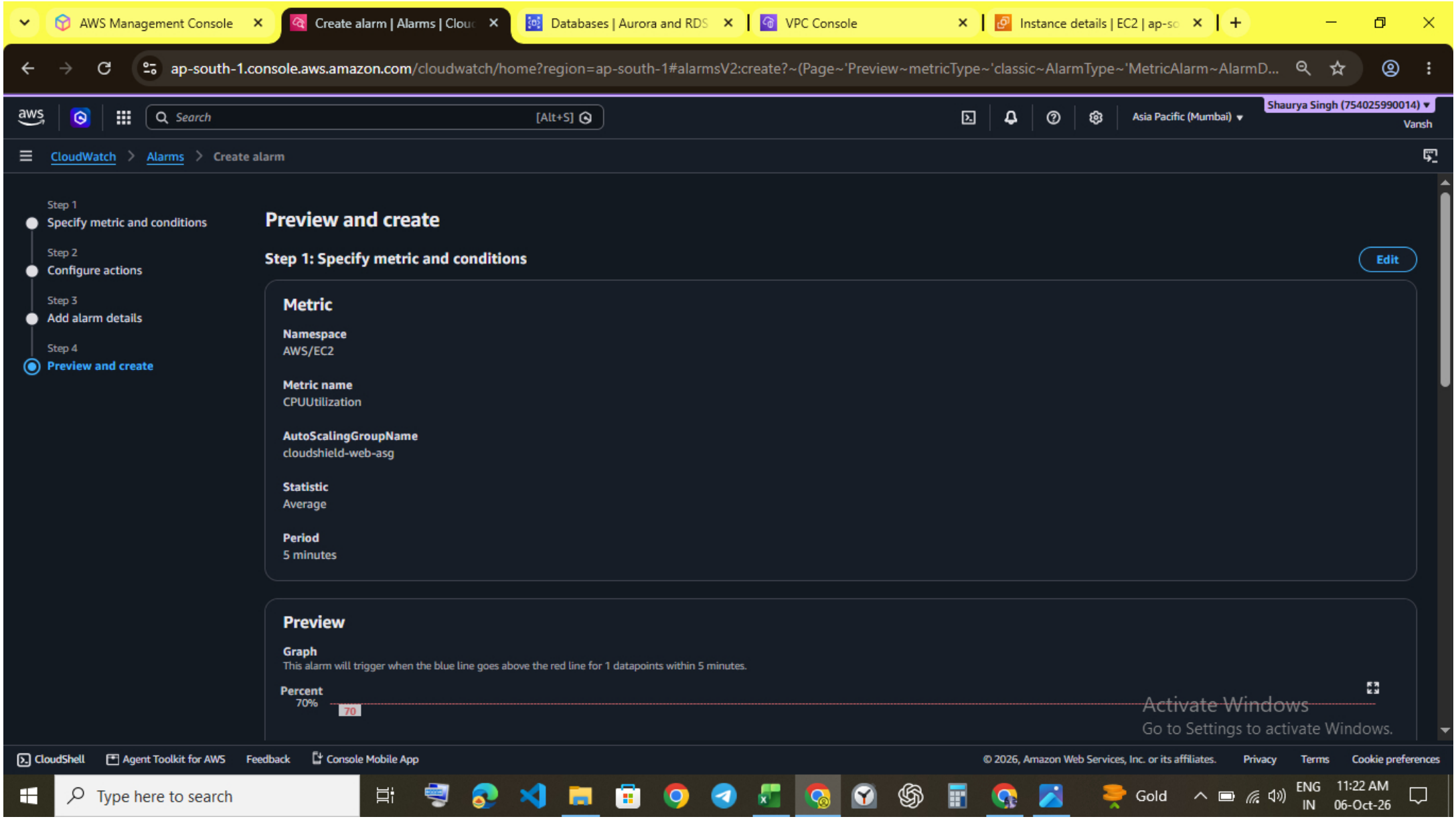
What was configured

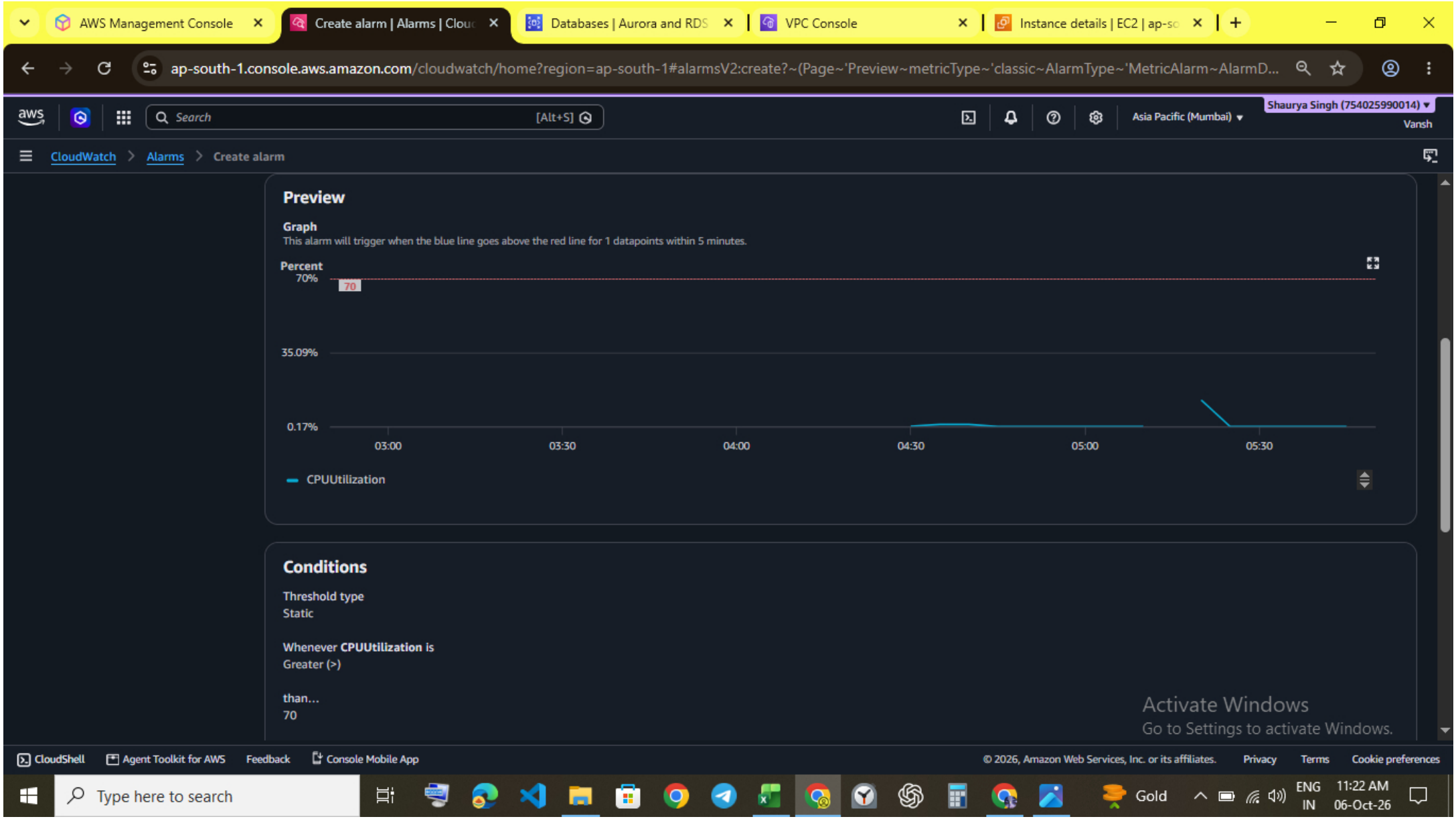
- CloudWatch monitoring/dashboard and the alarms shown in the architecture reference.
- SNS topic CloudWatch-alarms-topic with a confirmed email subscription.
- AWS Budget shown with a $50 monthly budget and thresholds at 50%, 80%, and 100%.

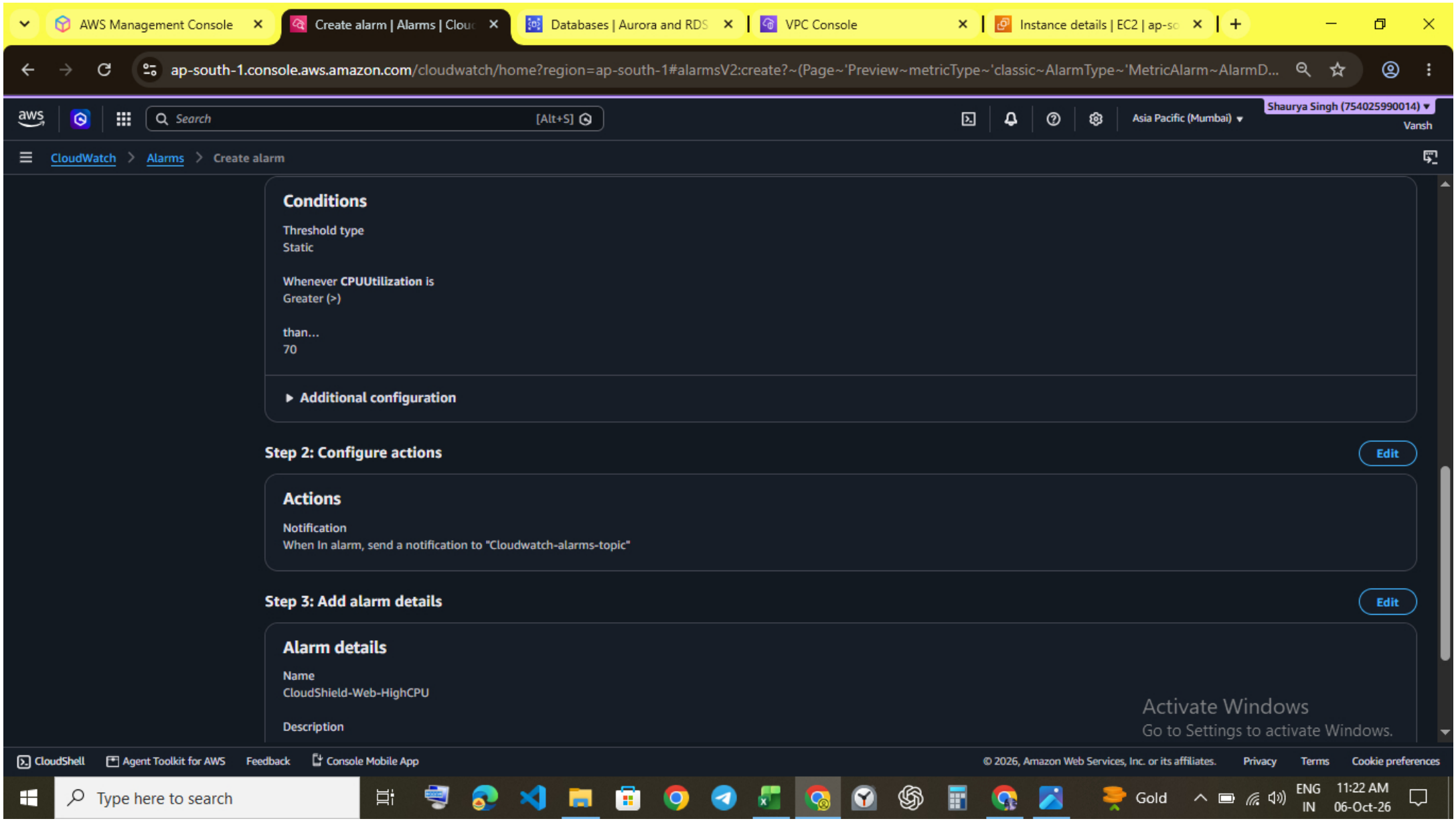
Operational checks

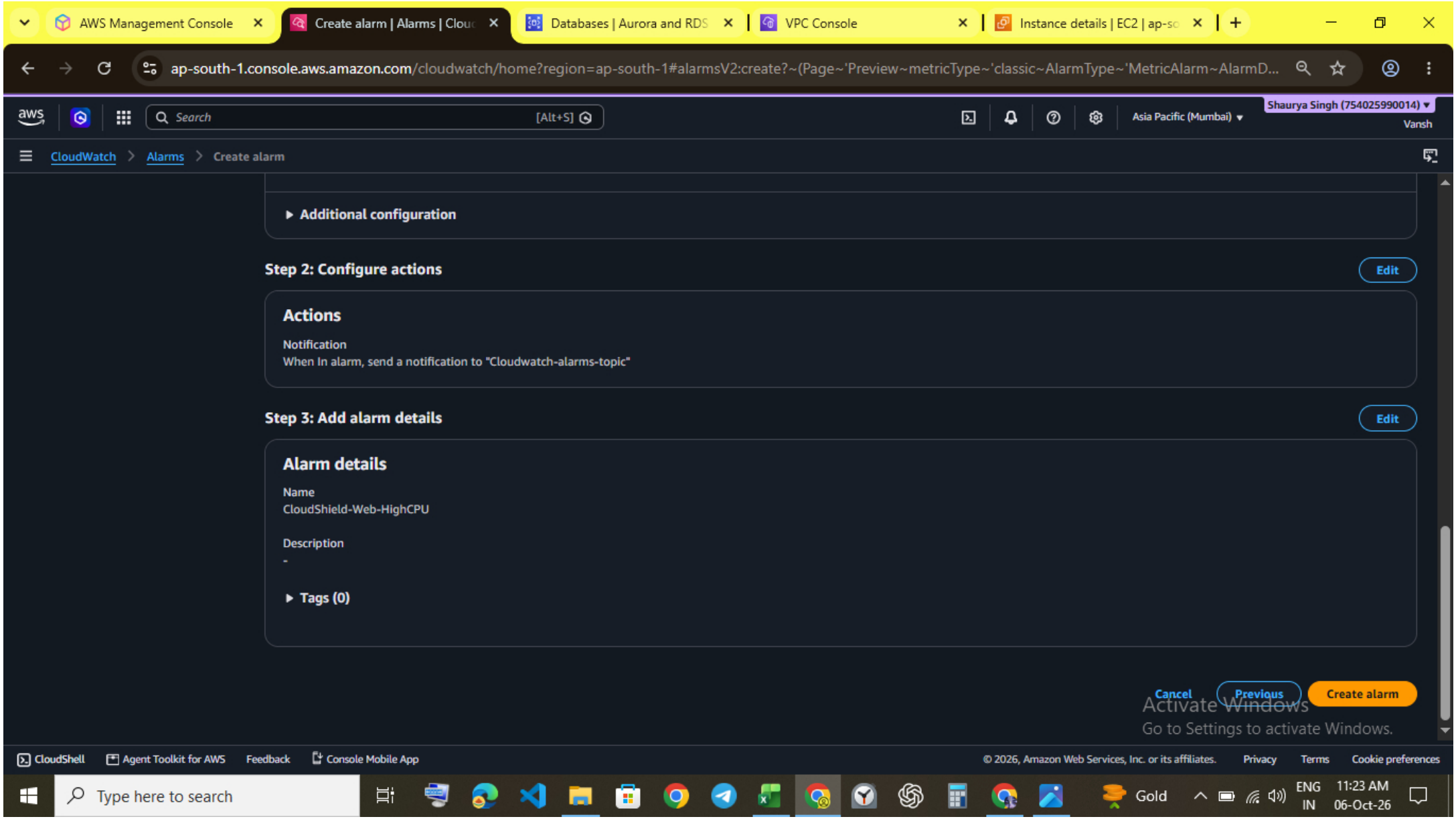
- Confirm each alarm’s metric, threshold, evaluation period, and SNS action in the console.
- Review unhealthy ALB targets and EC2/RDS metrics during troubleshooting.
- Confirm budget notification recipients and remember that a budget alert is not a spending limit.

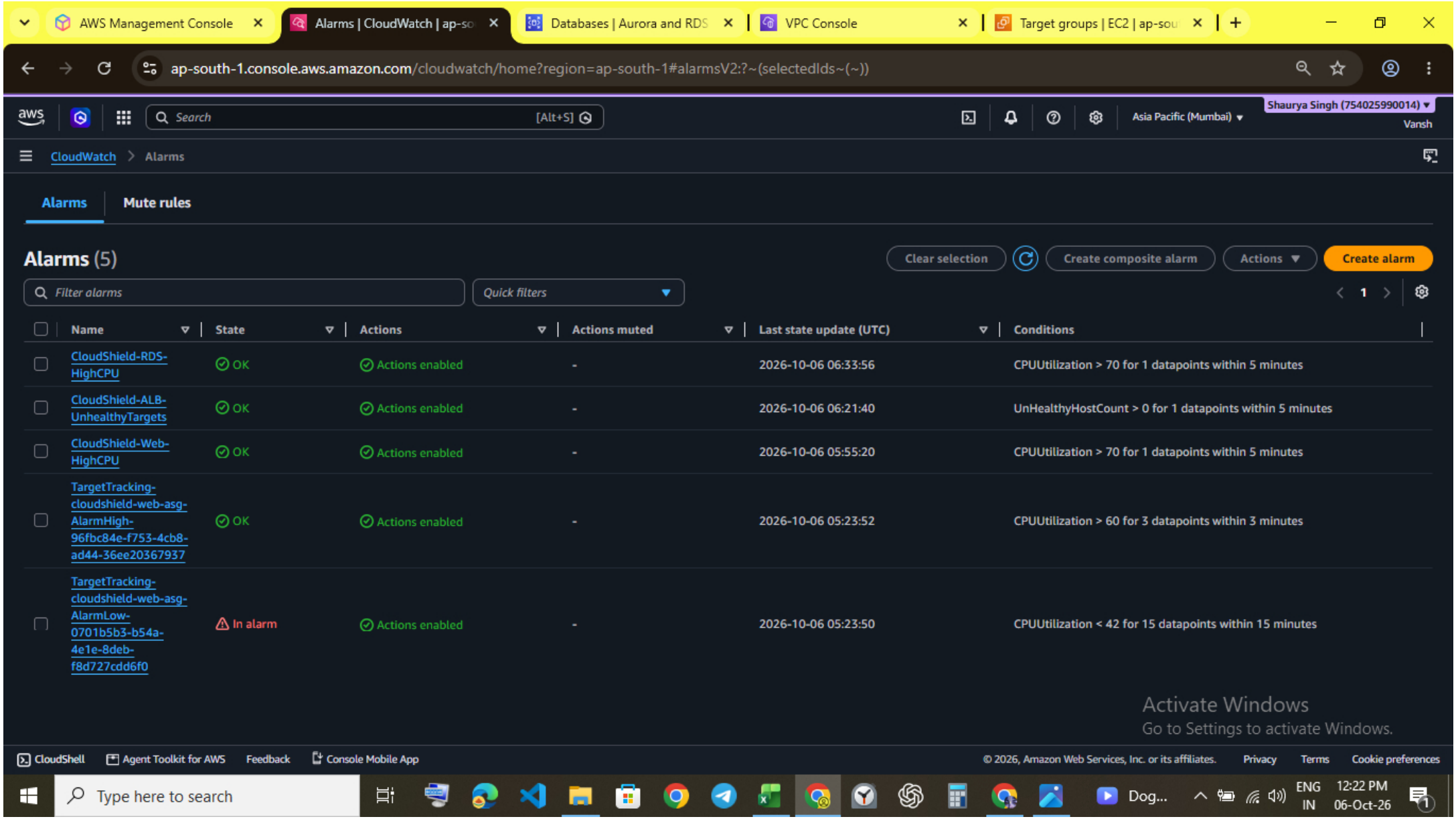
The original monitoring, notification, and budget evidence is preserved immediately after this phase.

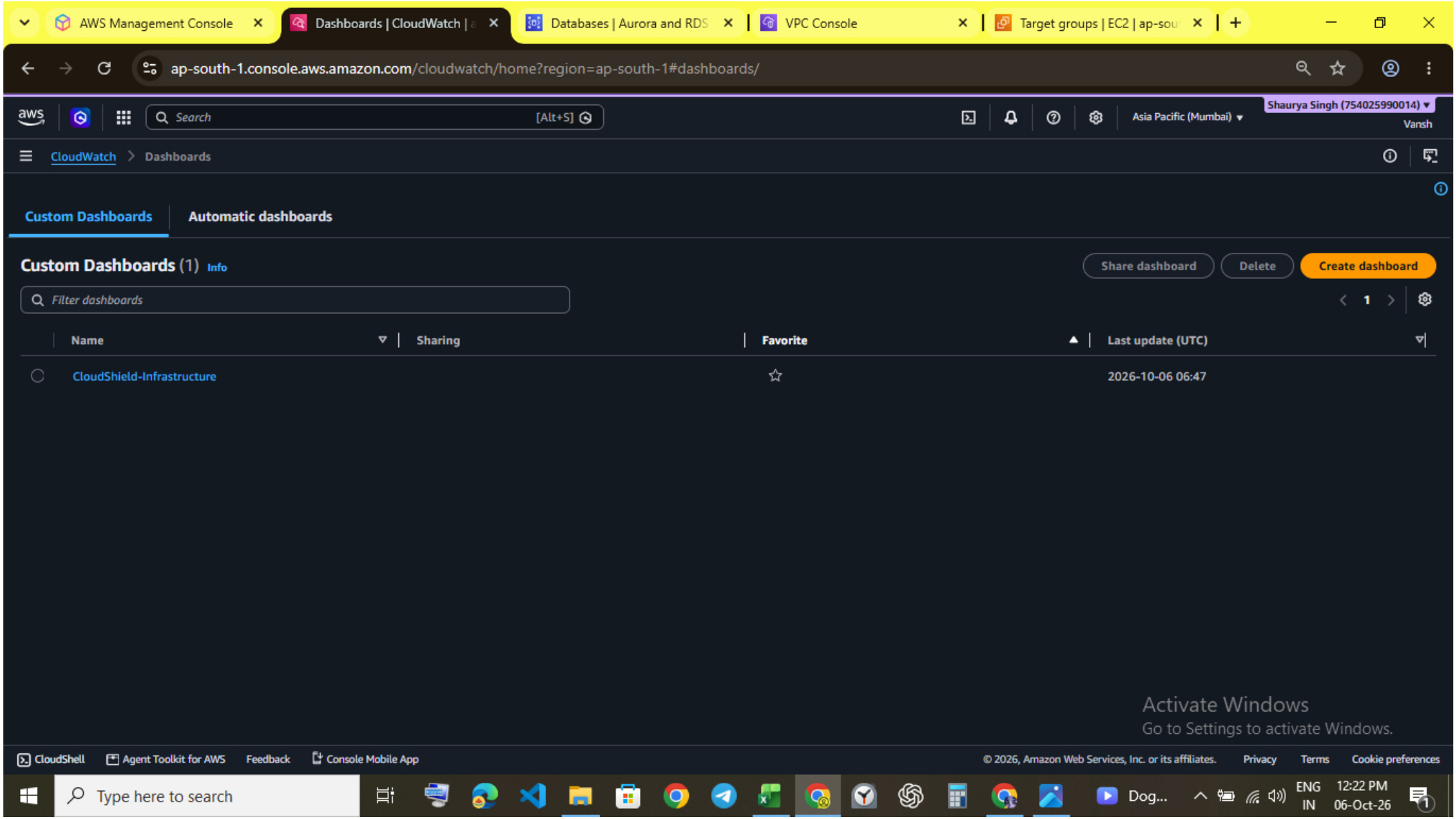

AWS Management Console
Create alarm | Alarms | Clou
Databases | Aurora and RDS
VPC Console
Instance details | EC2 | ap-so
ap-south-1.console.aws.amazon.com/cloudwatch/home?region=ap-south-1#alarmsV2:create?~(Page~'Preview~metricType~'classic~AlarmType~'MetricAlarm~AlarmD...
Search
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Asia Pacific (Mumbai)
Shaurya Singh (754025990014)
Vansh
CloudWatch
Alarms
Create alarm
Step 1
Specify metric and conditions
Step 2
Configure actions
Step 3
Add alarm details
Step 4
Preview and create
Preview and create
Step 1: Specify metric and conditions
Edit
Metric
Namespace
AWS/EC2
Metric name
CPUUtilization
AutoScalingGroupName
cloudshield-web-asg
Statistic
Average
Period
5 minutes
Preview
Graph
This alarm will trigger when the blue line goes above the red line for 1 datapoints within 5 minutes.
Percent
70%
70
Activate Windows
Go to Settings to activate Windows.
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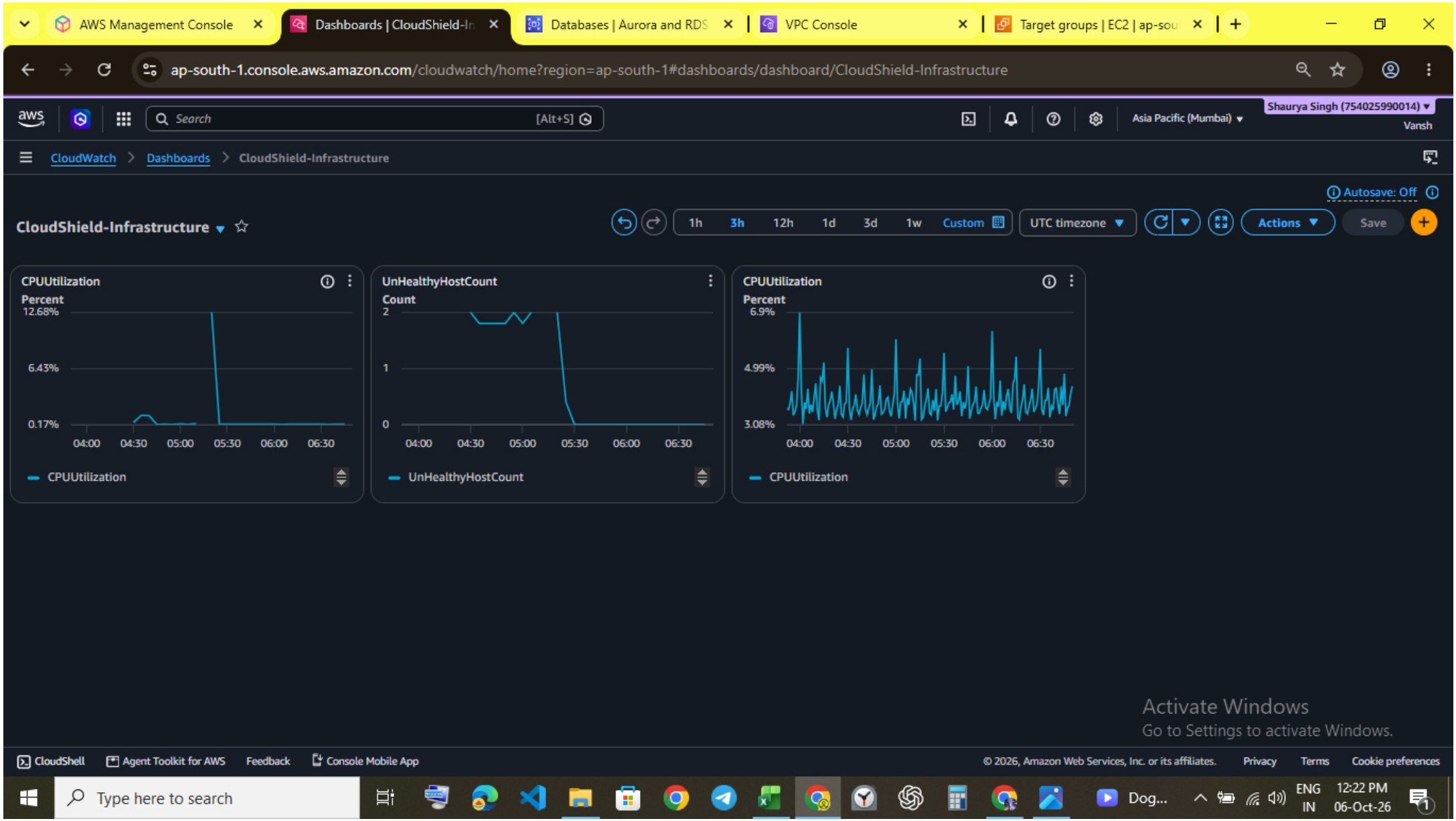

AWS Management Console
Create alarm | Alarms | Clou
Databases | Aurora and RDS
VPC Console
Instance details | EC2 | ap-so
ap-south-1.console.aws.amazon.com/cloudwatch/home?region=ap-south-1#alarmsV2:create?~(Page~'Preview~metricType~'classic~AlarmType~'MetricAlarm~AlarmD...
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Shaurya Singh (754025990014)
Vansh
CloudWatch
Alarms
Create alarm
Preview
Graph
This alarm will trigger when the blue line goes above the red line for 1 datapoints within 5 minutes.
Percent
70%
70
35.09%
0.17%
03:00
03:30
04:00
04:30
05:00
05:30
CPUUtilization
Conditions
Threshold type
Static
Whenever CPUUtilization is
Greater (>)
than...
70
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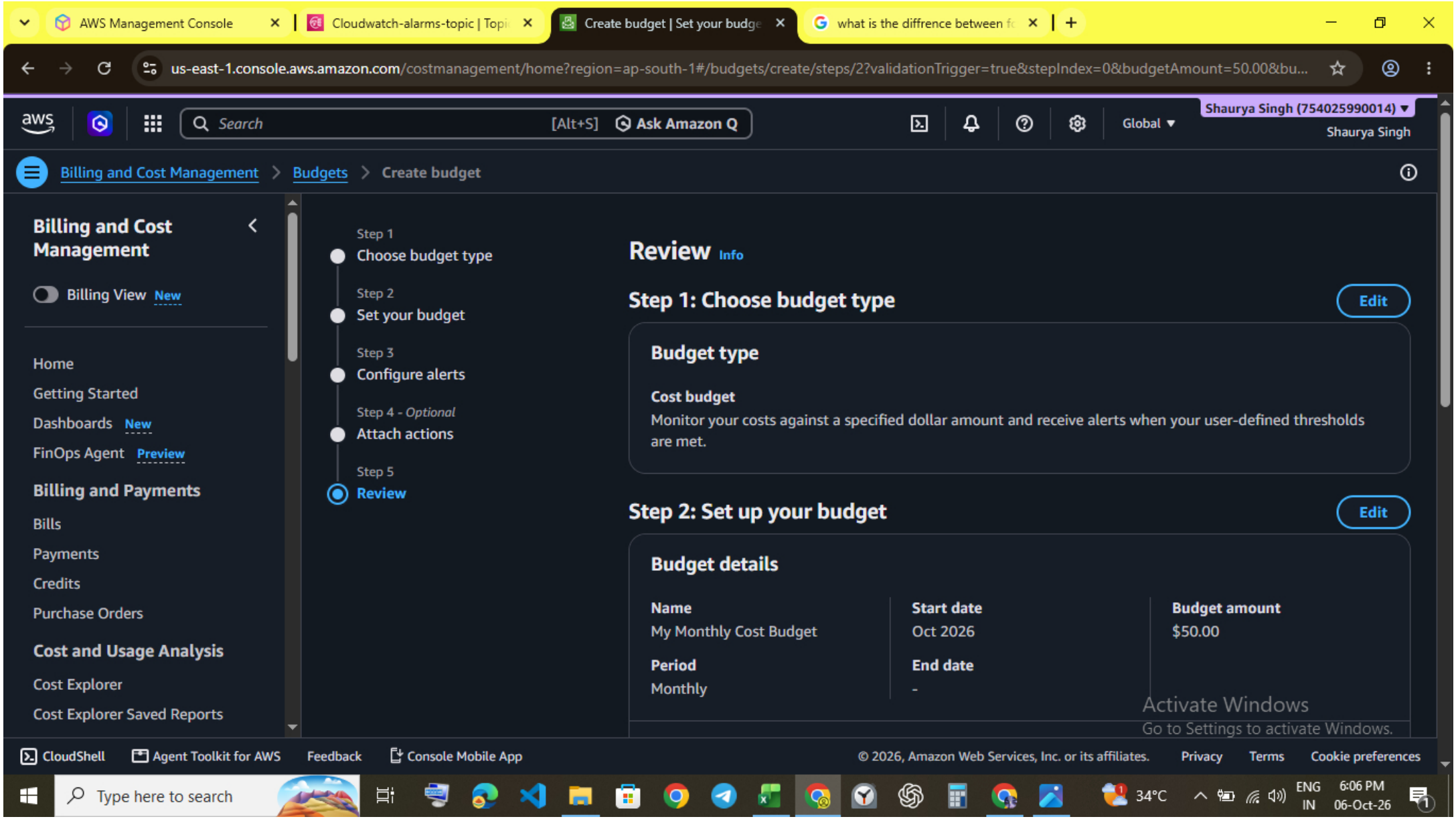

AWS Management Console
Create alarm | Alarms | Clou
Databases | Aurora and RDS
VPC Console
Instance details | EC2 | ap-so
ap-south-1.console.aws.amazon.com/cloudwatch/home?region=ap-south-1#alarmsV2:create?~(Page~'Preview~metricType~'classic~AlarmType~'MetricAlarm~AlarmD...
Search [Alt+S]
Asia Pacific (Mumbai)
Shaurya Singh (754025990014)
Vansh
CloudWatch > Alarms > Create alarm
Conditions
Threshold type
Static
Whenever CPUUtilization is
Greater (>)
than...
70
Additional configuration
Step 2: Configure actions
Edit
Actions
Notification
When In alarm, send a notification to "Cloudwatch-alarms-topic"
Step 3: Add alarm details
Edit
Alarm details
Name
CloudShield-Web-HighCPU
Description
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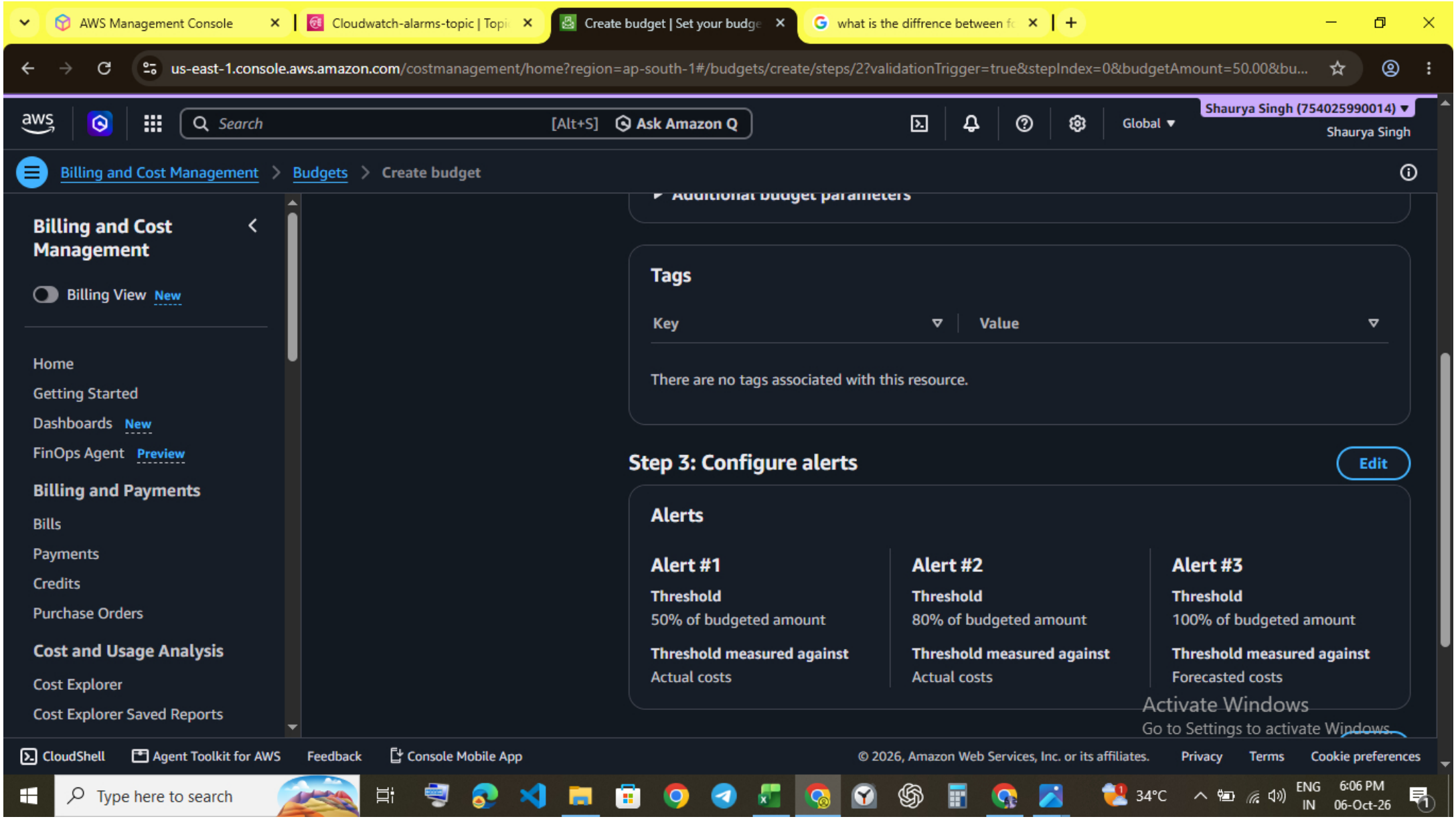

AWS Management Console
Create alarm | Alarms | Clou
Databases | Aurora and RDS
VPC Console
Instance details | EC2 | ap-so
ap-south-1.console.aws.amazon.com/cloudwatch/home?region=ap-south-1#alarmsV2:create?~(Page~'Preview~metricType~'classic~AlarmType~'MetricAlarm~AlarmD...
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CloudWatch
Alarms
Create alarm
Additional configuration
Step 2: Configure actions
Edit
Actions
Notification
When In alarm, send a notification to "Cloudwatch-alarms-topic"
Step 3: Add alarm details
Edit
Alarm details
Name
CloudShield-Web-HighCPU
Description
-
Tags (0)
Cancel
Previous
Create alarm
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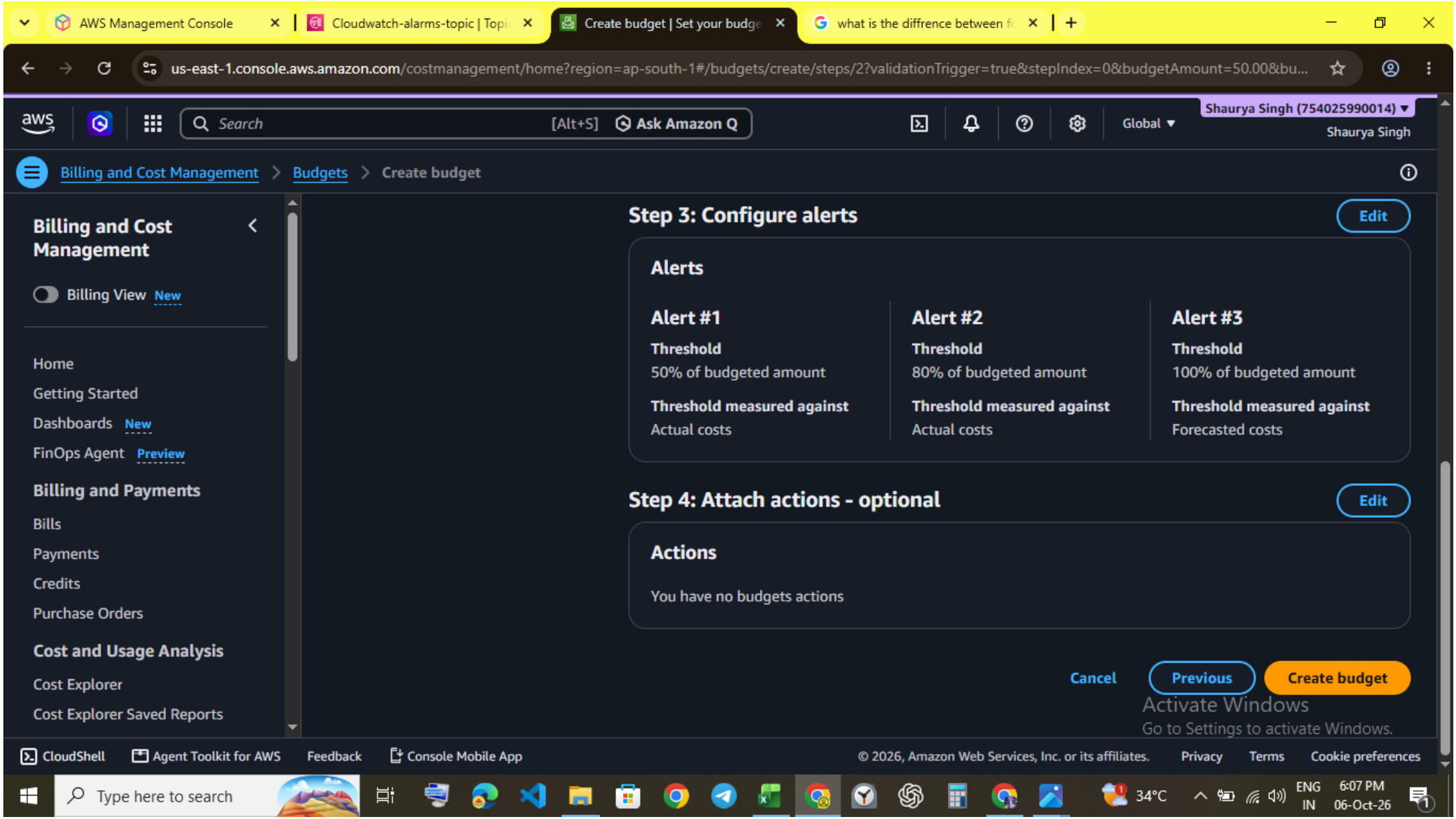

AWS Management Console
Alarms | CloudWatch | ap-so
Databases | Aurora and RDS
VPC Console
Target groups | EC2 | ap-sou
ap-south-1.console.aws.amazon.com/cloudwatch/home?region=ap-south-1#alarmsV2:?~(selectedIds~(~))
Search
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CloudWatch > Alarms
Alarms
Mute rules
Alarms (5)
Clear selection
Create composite alarm
Actions
Create alarm
Filter alarms
Quick filters
1
Name | State | Actions | Actions muted | Last state update (UTC) | Conditions
CloudShield-RDS-HighCPU | OK | Actions enabled | - | 2026-10-06 06:33:56 | CPUUtilization > 70 for 1 datapoints within 5 minutes
CloudShield-ALB-UnhealthyTargets | OK | Actions enabled | - | 2026-10-06 06:21:40 | UnHealthyHostCount > 0 for 1 datapoints within 5 minutes
CloudShield-Web-HighCPU | OK | Actions enabled | - | 2026-10-06 05:55:20 | CPUUtilization > 70 for 1 datapoints within 5 minutes
TargetTracking-cloudshield-web-asg-AlarmHigh-96fbc84e-f753-4cb8-ad44-36ee20367937 | OK | Actions enabled | - | 2026-10-06 05:23:52 | CPUUtilization > 60 for 3 datapoints within 3 minutes
TargetTracking-cloudshield-web-asg-AlarmLow-0701b5b3-b54a-4e1e-8deb-f8d727cdd6f0 | In alarm | Actions enabled | - | 2026-10-06 05:23:50 | CPUUtilization < 42 for 15 datapoints within 15 minutes
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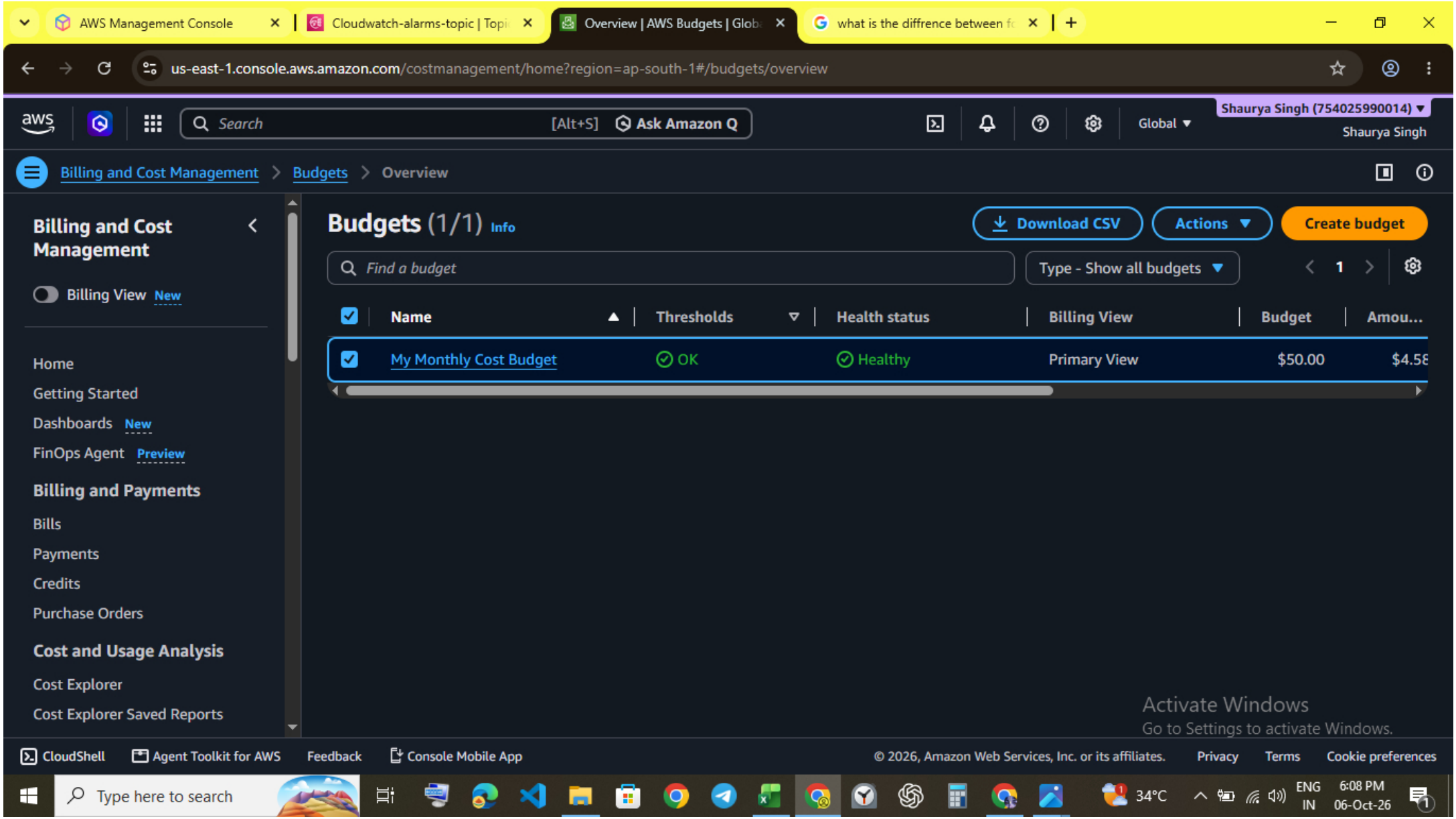

AWS Management Console
Dashboards | CloudWatch |
Databases | Aurora and RDS
VPC Console
Target groups | EC2 | ap-sou
ap-south-1.console.aws.amazon.com/cloudwatch/home?region=ap-south-1#dashboards/
Search
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CloudWatch
Dashboards
Custom Dashboards
Automatic dashboards
Custom Dashboards (1) Info
Share dashboard
Delete
Create dashboard
Filter dashboards
Name
Sharing
Favorite
Last update (UTC)
CloudShield-Infrastructure
2026-10-06 06:47
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AWS Management Console
Dashboards | CloudShield-In
Databases | Aurora and RDS
VPC Console
Target groups | EC2 | ap-sou
ap-south-1.console.aws.amazon.com/cloudwatch/home?region=ap-south-1#dashboards/dashboard/CloudShield-Infrastructure
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CloudWatch
Dashboards
CloudShield-Infrastructure
Autosave: Off
CloudShield-Infrastructure
1h
3h
12h
1d
3d
1w
Custom
UTC timezone
Actions
Save
CPUUtilization
Percent
12.68%
6.43%
0.17%
04:00
04:30
05:00
05:30
06:00
06:30
CPUUtilization
UnHealthyHostCount
Count
2
1
0
UnHealthyHostCount
CPUUtilization
Percent
6.9%
4.99%
3.08%
CPUUtilization
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AWS Management Console
Cloudwatch-alarms-topic | Topi
Create budget | Set your budge
what is the diffrence between f
us-east-1.console.aws.amazon.com/costmanagement/home?region=ap-south-1#/budgets/create/steps/2?validationTrigger=true&stepIndex=0&budgetAmount=50.00&bu...
Search
[Alt+S]
Ask Amazon Q
Global
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Billing and Cost Management
Budgets
Create budget
Billing and Cost Management
Billing View New
Home
Getting Started
Dashboards New
FinOps Agent Preview
Billing and Payments
Bills
Payments
Credits
Purchase Orders
Cost and Usage Analysis
Cost Explorer
Cost Explorer Saved Reports
Step 1
Choose budget type
Step 2
Set your budget
Step 3
Configure alerts
Step 4 - Optional
Attach actions
Step 5
Review
Review Info
Step 1: Choose budget type
Edit
Budget type
Cost budget
Monitor your costs against a specified dollar amount and receive alerts when your user-defined thresholds are met.
Step 2: Set up your budget
Edit
Budget details
Name
My Monthly Cost Budget
Start date
Oct 2026
Budget amount
$50.00
Period
Monthly
End date
-
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Go to Settings to activate Windows.
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AWS Management Console
Cloudwatch-alarms-topic | Topi
Create budget | Set your budge
what is the diffrence between f
us-east-1.console.aws.amazon.com/costmanagement/home?region=ap-south-1#/budgets/create/steps/2?validationTrigger=true&stepIndex=0&budgetAmount=50.00&bu...
aws
Search
[Alt+S]
Ask Amazon Q
Global
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Billing and Cost Management > Budgets > Create budget
Billing and Cost Management
Billing View New
Home
Getting Started
Dashboards New
FinOps Agent Preview
Billing and Payments
Bills
Payments
Credits
Purchase Orders
Cost and Usage Analysis
Cost Explorer
Cost Explorer Saved Reports
Additional budget parameters
Tags
Key
Value
There are no tags associated with this resource.
Step 3: Configure alerts
Edit
Alerts
Alert #1
Threshold
50% of budgeted amount
Threshold measured against
Actual costs
Alert #2
Threshold
80% of budgeted amount
Threshold measured against
Actual costs
Alert #3
Threshold
100% of budgeted amount
Threshold measured against
Forecasted costs
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Cloudwatch-alarms-topic | Topi
Create budget | Set your budge
what is the diffrence between f
us-east-1.console.aws.amazon.com/costmanagement/home?region=ap-south-1#/budgets/create/steps/2?validationTrigger=true&stepIndex=0&budgetAmount=50.00&bu...
aws
Search
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Ask Amazon Q
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Billing and Cost Management > Budgets > Create budget
Billing and Cost Management
Billing View New
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Dashboards New
FinOps Agent Preview
Billing and Payments
Bills
Payments
Credits
Purchase Orders
Cost and Usage Analysis
Cost Explorer
Cost Explorer Saved Reports
Step 3: Configure alerts
Edit
Alerts
Alert #1
Threshold
50% of budgeted amount
Threshold measured against
Actual costs
Alert #2
Threshold
80% of budgeted amount
Threshold measured against
Actual costs
Alert #3
Threshold
100% of budgeted amount
Threshold measured against
Forecasted costs
Step 4: Attach actions - optional
Edit
Actions
You have no budgets actions
Cancel
Previous
Create budget
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AWS Management Console
Cloudwatch-alarms-topic | Topi
Overview | AWS Budgets | Glob
what is the diffrence between f
us-east-1.console.aws.amazon.com/costmanagement/home?region=ap-south-1#/budgets/overview
Search
[Alt+S]
Ask Amazon Q
Global
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Billing and Cost Management
Budgets
Overview
Billing and Cost Management
Billing View New
Home
Getting Started
Dashboards New
FinOps Agent Preview
Billing and Payments
Bills
Payments
Credits
Purchase Orders
Cost and Usage Analysis
Cost Explorer
Cost Explorer Saved Reports
Budgets (1/1) Info
Download CSV
Actions
Create budget
Find a budget
Type - Show all budgets
1
Name
Thresholds
Health status
Billing View
Budget
Amou...
My Monthly Cost Budget
OK
Healthy
Primary View
$50.00
$4.58
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